RESEARCH IN SCIENCE & TECHNOLOGICAL EDUCATION

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NEW ARTICLE

Automated assessment of the nature of scientific inquiry (NOSI) using machine learning and transformer-based language models

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WHAT DID THE RESEARCHERS INVESTIGATE?

- Hand-scoring open-ended instruments such as the Views About Scientific Inquiry (VASI) is resource-intensive.
- The authors developed and validated a machine learning framework to automate scoring of responses from 503 Turkish pre-service teachers.
- They compared two approaches: a feature-based Random Forest model and a transformer-based BERTurk model, using 5-fold cross-validation.

Mesci et al. (2026) · Research in Science & Technological Education · doi.org/10.1080/02635143.2026.2731035

WHAT DID THEY FIND?

- The Random Forest model gave a reliable, stable baseline that aligned strongly with expert scores.
- BERTurk looked promising initially, but its performance collapsed non-reproducibly in stability tests, highlighting fragility with limited data and a morphologically complex language.
- The authors argue that stability and interpretability must take precedence over peak performance.

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